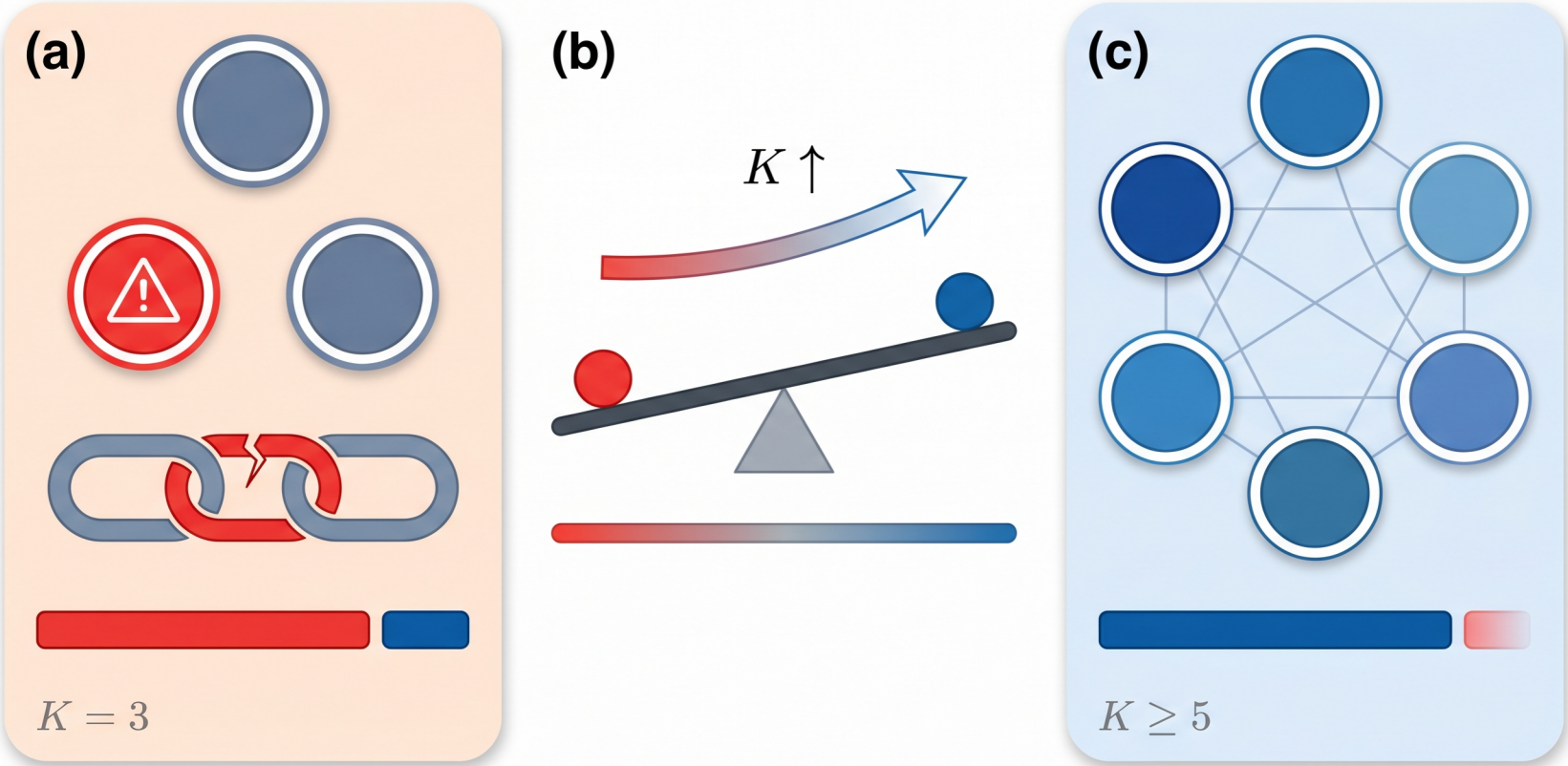

(a)
$K = 3$
(b)
$K \uparrow$
(c)
$K \geq 5$